

LeCamera: Dynamic and Expressive Human Tracking System With Computer Vision and Low-cost Robotics

Qihan Jiang
New York University
qj351@nyu.edu

Bairui Su
New York University
bs4396@nyu.edu

Abstract

Documenting the process of developing interactive art projects such as kinetic installations and immersive environments is significant for artists and creative technologies. The traditional documenting process also includes hiring a team to film the process, which is inefficient and expensive, or using one or several static cameras for filming, which not only provide limited perspective and angle, but also constrain the movement of the creator, since they might move outside of the viewfinder and thus not being recorded. In this project, we introduce LeCamera, a dynamic human tracking system powered by affordable robotics and computer vision techniques. We adopted SO-101 robot as a camera tracker, and compared the performances of building such system using object segmentation and inverse kinematics and end-to-end training. The system provides promising outcome for conducting human tracking in a create project development scenerio.

1. Github Link

<https://github.com/jqhisme/LeCamera>

2. Introduction

Documenting the process of developing interactive art projects—such as kinetic installations, performative sculptures, and immersive environments—plays a crucial role in both artistic reflection and the dissemination of creative technologies. Unlike static artworks, these projects often evolve through embodied interaction, spatial experimentation, and iterative prototyping. Capturing such processes in a meaningful way requires visual documentation that is flexible, continuous, and responsive to the creator’s movement in space.

Traditional documentation practices, however, present significant limitations. Professional filming typically involves hiring a dedicated crew, which is costly and impractical for long-term or iterative creative development. Alter-

natively, artists often rely on one or more static cameras, which provide limited viewpoints and constrain the creator’s movement: once the artist leaves the camera’s field of view, the documentation fails. These constraints not only reduce the richness of recorded perspectives but can also interfere with the creative process itself, discouraging natural movement and spatial exploration.

To address these challenges, we introduce LeCamera, a dynamic human-tracking camera system designed for documenting creative project development. LeCamera leverages affordable robotics and computer vision techniques to autonomously track a human subject and generate diverse visual perspectives without requiring a camera operator. The system is built on the SO-101 robot platform, repurposed as a mobile camera tracker capable of following a creator’s motion in real time while maintaining expressive and adaptive camera trajectories.

We explore two approaches for enabling human tracking and camera control. The first approach combines YOLO-based human detection and segmentation with inverse kinematics (IK) to analytically compute camera movements. This pipeline is straightforward to implement and computationally efficient, making it suitable for rapid prototyping. However, the resulting camera behavior tends to be deterministic and mechanically constrained, producing trajectories that lack diversity and appear unnatural in expressive documentation contexts.

The second approach investigates imitation learning, using an Action Chunking Transformer (ACT) policy trained end-to-end to map visual observations directly to camera control actions. By learning from human demonstrations, this method produces more varied, human-like camera movements and yields richer perspectives that better reflect the dynamics of creative practice. While this approach shows promising expressive potential, it remains less robust and more challenging to generalize across environments and creators, highlighting an important trade-off between controllability and expressiveness.

Through LeCamera, we aim to answer the following question: How can we design a more natural and expres-

sive object tracking system for creative project documentation that produces diverse visual perspectives while preserving the creator’s freedom of movement? By comparing analytical and learning-based control strategies, this project contributes insights into the design of autonomous camera systems at the intersection of computer vision, robotics, and interactive art practice.

3. Related Work

3.1. Expressive Robotics

Research in expressive robotics investigates how motion can convey intent, attention, and affect in non-anthropomorphic systems. *LeLamp*, an open-source robotic lamp developed by the Human–Computer Interaction Lab and inspired by Apple’s *Elegant* animation principles, demonstrates how carefully designed kinematic motion can produce legible and expressive behavior. Similarly, *Shoggoth Mini* explores organic and creature-like movement using a compact robotic structure, emphasizing expressiveness over task efficiency. These works motivate our treatment of camera motion as an expressive element in creative documentation rather than a purely functional tracking behavior.

3.2. Imitation Learning for Control

Imitation learning enables the acquisition of control policies directly from human demonstrations, allowing end-to-end mappings from perception to action. Recent approaches such as the Action Chunking Transformer (ACT) model long-horizon action sequences and have shown promise in generating diverse and human-like trajectories. Open-source frameworks including the *LeRobot* library support human-in-the-loop data collection and deployment of imitation learning policies on real robotic systems. While imitation learning offers increased expressiveness compared to analytical controllers, these methods often face challenges in robustness and generalization.

4. Experiments

We use a SO-101 affordable robotics arm for conducting s. A standard webcam is attached to the robotic arm. We used LeRobot library for controlling the robot.

4.1. YOLO + Inverse Kinematics

Our first experiment uses computer vision model to identify the location of the human, and then use Inverse Kinematics. The implementation of the Inverse Kinematics is adapted from the [XLeRobot](#) Library. To make sure the robot can move smoothly and avoid any choppy behavior, we use yolo11s, a relatively small model. This model allows quick inference while still maintaining high accuracies in iden-



Figure 1. LeCamera using YOLO and IK following people

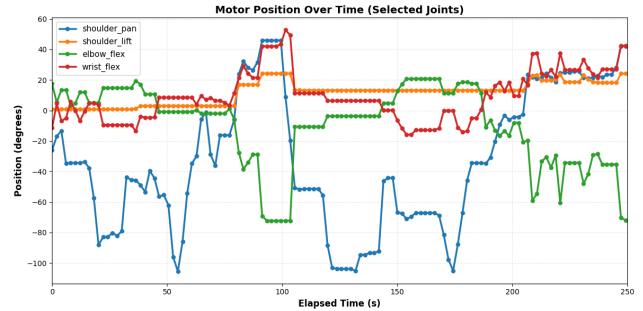


Figure 2. Movement of each joint over time

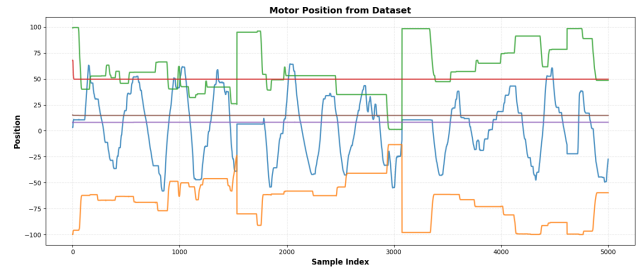


Figure 3. The collected data of motor movement

tifying people. Using this model allows smooth tracking running on a machine with one NVIDIA 4070 GPU.

4.2. Imitation Learning

4.2.1. Data Collection

We collect 20 episodes of data. We recorded the movement of each motor, as well as the camera input for each timestamp. The data is recorded using a leader arm and a follower arm setup, where one person moves the leader arm and follow the other person manually, and the follower arm follows the behavior. We recorded people doing different task one may encounter when making an physical interactive project, including moving objects, running, and walking. The dataset is recorded using LerobotDataset and being pushed to the huggingface hub for training.

Comparing the graph of motor movement by the inverse kinematics and the movement controlled by a human, we can observe differences in movement patterns. The IK movement produce more stiff curves, when the yolo model

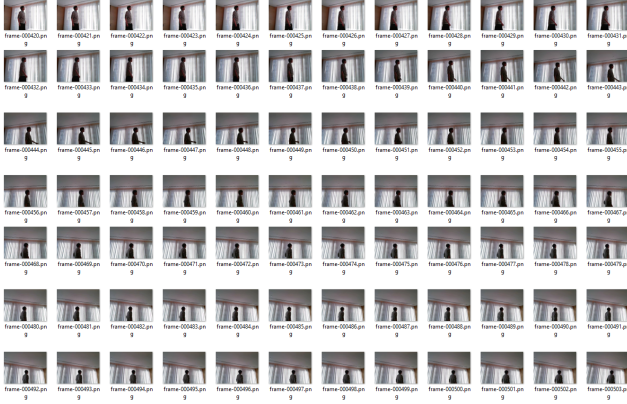


Figure 4. Camera Images from the Manually Collected Dataset



Figure 5. Loss during training

identifies a person and the mechanism immediate turn the robot toward it, while in the human controlled version, curves are more smooth with more gradual increase and decrease. Additionally, the human controlled patterns also reveals some regularities and relationship between motors, showing that humans may prefer to control the camera in some specific way. These two observation shows that imitation learning for human tracking will be helpful for machines to record smooth and natural footage.

4.2.2. Training Policies

We trained an Action Chunking Transformer (ACT) policy and a SmolVLM policy. Both of the policies are trained on 10,000 steps. We used Adam as the optimizer with a learning rate decay. We trained the model using one NVIDIA RTX 6000 GPU. It takes about 2 hours for each policy to be trained. The loss curve shows that SmolVLM is better at learning the dataset then the ACT policy, this is probably due to the fact that SmolVLM base is trained on vision language datasets, thereby having better vision understanding.

4.2.3. Policy Evaluation

We evaluated the policy by human evaluation, motor path visualization, and the percentage of frames recorded by the robot having a human in it, given a human is presented in the area that it can capture. We recorded the data with 5 episodes, and 200 steps max for each episode.



Figure 6. ACT motor position during evaluation

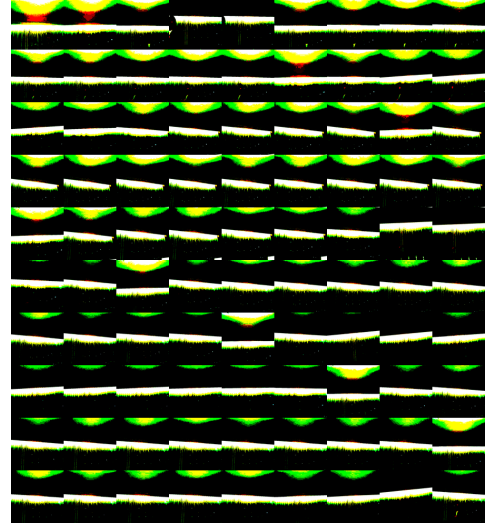


Figure 7. Images captured by ACT policy

The ACT policy is not yielding useful results. When running on this policy, the robot action seems to be collapsed into following a same path, regardless of where the human is presented. The motor position visualization also support this. Since the evaluation run contains 5 episode, the trajectories of the motors are similar across 5 episodes, with a sudden change at the start of each episode. This is probably due to, in the recorded dataset, we tend to move arm most drastically at the start of the episode to point the camera toward the target. Among all the images captured by this policy, less than 10 out of 90 contain the person.

The SmolVLM policy shows promising result of tracking a person. Among 90 images captured by this policy, most of them contain the human. Nevertheless, in many captures, the person is not being place in the center. From human observation, we observed that the policy cannot follow the person as smooth as the inverse kinematics strategy. It takes some time to explore where the person is and slowly move toward the person's location. In the scenario when the person is moving quickly, this policy cannot capture the person. Although the motor movement also has some signs of repetitiveness across different episodes, they seem much



Figure 8. Images captured by smolVLM

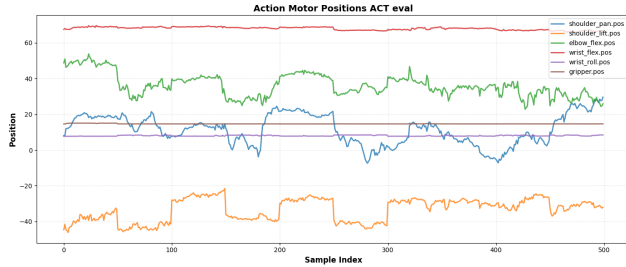


Figure 9. Motor movement from the smolVLM policy evaluation

less obvious than the ACT policy and more dependent to the camera input.

5. Conclusion

In this project, we studies how computer vision and robotics and assist artists and creators to document their process of developing physical, usually large scale, installation or environment-like works. We investigated two strategies. The first strategy uses object detection model and inverse kinematics for control, where we uses YOLO as the object detection model. The second strategy uses end to end training. We collect data of 20 episodes of human tracking using a leader arm and follower arm setup, and then we trained two policies using Action Chunking Transformers (ACT) and smolVLM. These policies show a more diverse and smooth movement comparing to the Inverse Kinematics control, but lack accuracy in tracking people. Additionally, due to the limited location of where the data are collected, these policies is bot robust enough to generalize to different locations.

For the future works, building a more diverse and comprehensive dataset will be helpful. Current size of the

dataset (20 episodes) is very small, and thus very hard to yield outcomes that are generalizable.

Additionally, end to end training does not explicitly provide information of following an object. In our future works, we plan to explore how we an inject the information specifying what object to track, potentially aggregating the binary segmentation image or feature map from yolo in the observation space. Finally, the processes of work documentation of such physical project is generally include more than tracking humans. Artist may desire more diverse focuses such as tracking the object, the artwork, the processes of one talking to each other, and more. In order to accomplish this, building a system that integrates high level planning (using and large vision language model) and low level control (via close loop or open loop policies) will be helpful. This will allow more flexible control of the tracking system.

6. LLM Usage Statement

We used LLM for revising code and improve writing. We are responsible for all the result we provides.

7. Equal Contribution Statement

Qihan Jiang and Bairui Su make equal contribution to the project. We conducted research, data collection, training, and evaluation together.

8. References

- [1] Hu, Y., Huang, P., Sivapurapu, M., and Zhang, J. 2025. ELEGNT: Expressive and Functional Movement Design for Non-Anthropomorphic Robot. arXiv preprint arXiv:2501.12493.
- [2] HuggingFace. 2024. LeRobot: State-of-the-art Robotics Learning with PyTorch. <https://github.com/huggingface/lerobot>.
- [3] Lecarme, M. 2023. Shoggoth Mini. <https://www.matthieulc.com/posts/shoggoth-mini/>.
- [4] Marafioti, A., Zohar, O., Farré, M., Noyan, M., Bakouch, E., Cuenca, P., Zakka, C., Allal, L. B., Lozhkov, A., Tazi, N., Srivastav, V., Lochner, J., Larcher, H., Morlon, M., Tunstall, L., von Werra, L., and Wolf, T. 2025. SmolVLM: Redefining small and efficient multimodal models. arXiv preprint arXiv:2504.05299.
- [5] Zhao, T. Z., Kumar, V., Levine, S., and Finn, C. 2023. Learning Fine-Grained Bimanual Manipulation with Low-Cost Hardware. arXiv preprint arXiv:2304.13705.